In [6]:
```
import pandas as pd

df = pd.read_csv('RetailCleaned.csv', header=None)
df.columns = ['Invoice' , 'StockCode' , 'Description' , 'Quantity' , 'InvoiceDate'

df['InvoiceDate'] = pd.to_datetime(df['InvoiceDate'])
print(df.shape)
df.info()
```

```
C:\Users\Arena\AppData\Local\Temp\ipykernel_21320\1293971194.py:3: DtypeWarning: Col
umns (0: 0) have mixed types. Specify dtype option on import or set low_memory=Fals
e.
  df = pd.read_csv('RetailCleaned.csv', header=None)
(1033036, 9)
<class 'pandas.DataFrame'>
RangeIndex: 1033036 entries, 0 to 1033035
Data columns (total 9 columns):
 #   Column       Non-Null Count    Dtype
---  ------       --------------    -----
 0   Invoice      1033036 non-null  object
 1   StockCode    1033036 non-null  str
 2   Description  1028761 non-null  str
 3   Quantity     1033036 non-null  int64
 4   InvoiceDate  1033036 non-null  datetime64[ns]
 5   Price        1033036 non-null  float64
 6   Customer_ID  797885 non-null   float64
 7   Country      1033036 non-null  str
 8   SourceSheet  1033036 non-null  str
dtypes: datetime64[ns](1), float64(2), int64(1), object(1), str(4)
memory usage: 70.9+ MB
```

In [7]:
```
#Filter And Calculate Recency , Frequency , Monetary

rfm_data = df[(df['Customer_ID'].notna()) & (df['Quantity'] > 0)].copy()
snapshot_date = pd.to_datetime('2011-12-10')

rfm = rfm_data.groupby('Customer_ID').agg(
      Recency=('InvoiceDate' , lambda x: (snapshot_date - x.max()).days),
      Frequency=('Invoice' , 'nunique'),
      Monetary=('Quantity' , lambda x: (x * rfm_data.loc[x.index, 'Price']).sum())
).reset_index()
```

In [8]:
```
#Score Customers and Assign Business Segments

rfm['R_Score'] = pd.qcut(rfm['Recency'], 5, labels=[5, 4, 3, 2, 1]).astype(int)
rfm['F_Score'] = pd.qcut(rfm['Frequency'].rank(method='first'), 5, labels=[1, 2, 3,
rfm['M_Score'] = pd.qcut(rfm['Monetary'], 5, labels=[1, 2, 3, 4, 5]).astype(int)

def segment_customer(row):
    if row['R_Score'] >= 4 and row['F_Score'] >= 4 and row['M_Score'] >= 4:
        return 'Champions'
    elif row['R_Score'] >= 3 and row['F_Score'] >= 3:
        return 'Loyal'
    elif row['R_Score'] <= 2 and row['F_Score'] >= 3:
```

```
        return 'At Risk'
    elif row['R_Score'] >= 4 and row['F_Score'] <= 2:
        return 'New'
    elif row['R_Score'] == 3 and row['F_Score'] <= 2:
        return 'Needs Attention'
    else:
        return 'Lost'

rfm['Segment'] = rfm.apply(segment_customer, axis=1)
rfm['IsChurned'] = rfm['Recency'] > 90
```

In [9]:
```
!pip install openpyxl
```

```
Requirement already satisfied: openpyxl in .\AppData\Local\Programs\Python\Python313
\Lib\site-packages (3.1.5)
Requirement already satisfied: et-xmlfile in .\AppData\Local\Programs\Python\Python3
13\Lib\site-packages (from openpyxl) (2.0.0)
[notice] A new release of pip is available: 26.2.1 -> 26.2.2
[notice] To update, run: python.exe -m pip install --upgrade pip
```

In [10]:
```
#Merge RFM Results Back Onto Transactions and Export

df = df[(df['Customer_ID'].notna()) & (df['Quantity'] > 0)].copy()
df['InvoiceMonth'] = df['InvoiceDate'].dt.to_period('M').dt.to_timestamp()
df['Revenue'] = df['Quantity']*df['Price']

df_final = df.merge(
    rfm[['Customer_ID' , 'Recency' , 'Frequency' , 'Monetary' , 'R_Score' , 'F_Scor
    on = 'Customer_ID' , how = 'left'

)

df_final.to_excel('EcommerceDashboardData.xlsx', index=False)
```

In [ ]: